

Relate graphically the 3 notions below:

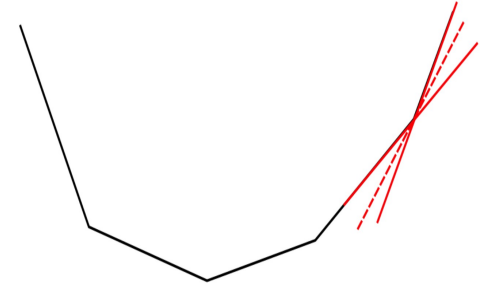
- 1) gradient
- 2) sub-gradient
- 3) optimum of a convex function

Fixing Bugs...

Let for all x : $f(x) \geq f(x^0) + \langle g, x - x^0 \rangle$

g – subgradient of f in $x^0 \in \text{dom} f$.

Proposition: ∂f – convex set



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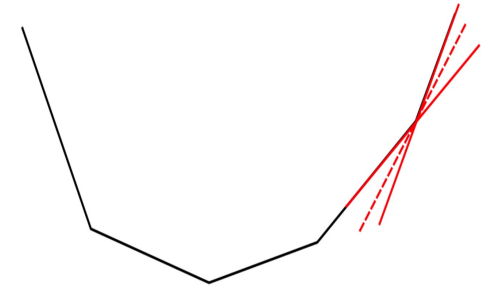
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Proof:

$$f(x) \geq f(x^0) + \langle g, x - x^0 \rangle$$

$$f(x) \geq f(x^0) + \langle g', x - x^0 \rangle$$



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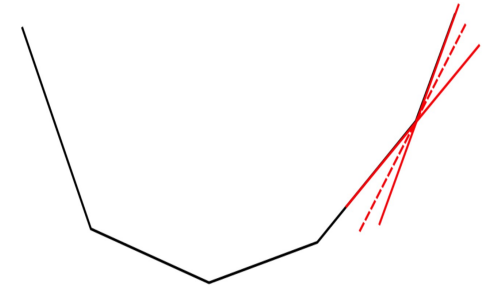
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$$f(x) \geq f(x^0) + \langle g, x - x^0 \rangle$$

$$f(x) \geq f(x^0) + \langle g', x - x^0 \rangle$$

$$\alpha f(x) + (1 - \alpha)f(x) = f(x)$$

$$\begin{aligned} &\geq \alpha f(x^0) + (1 - \alpha)f(x^0) + \langle \alpha g, x - x^0 \rangle + \langle (1 - \alpha)g', x - x^0 \rangle \\ &= f(x^0) + \langle \alpha g + (1 - \alpha)g', x - x^0 \rangle \end{aligned}$$



Proposition: ∂f – convex set

Proposition Hypothesis:

$\tilde{f}(y) = \max_{x \in D} f(x, y)$ f – differentiable on y , D – arbitrary set

Then $\partial \tilde{f}(y) = \{\nabla_y f(x^*(y), y) : x^*(y) \in \arg \max_{x \in D} f(x, y)\}$ (*)

What does holds for the set defined in (*):

- 1) it might be non-convex
- 2) it is always convex
- 3) No idea

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What if D is discrete?

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What if: $\tilde{f}(y) = \begin{cases} y^2, & x = 1 \\ y, & x = 2 \end{cases}$

Proposition: ∂f – convex set

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Proposition: ∂f – convex set

Proposition:

$\tilde{f}(y) = \max_{x \in D} f(x, y)$ f – differentiable on y , D – arbitrary set

Then $\partial \tilde{f}(y) = \text{conv} \{ \nabla_y f(x^*(y), y) : x^*(y) \in \arg \max_{x \in D} f(x, y) \}$ (*)

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Exercise: $\tilde{f}(y) = \begin{cases} y^2, & x = 1 \\ y, & x = 2 \end{cases} \quad \partial \tilde{f}(y) \supseteq \{2y, 1\}$

Check by definition that any element of $\text{conv}\{2y, 1\}$ is indeed a sub-gradient .

Exercise: Prove (*)

Recall: Super-gradient for the Dual LP

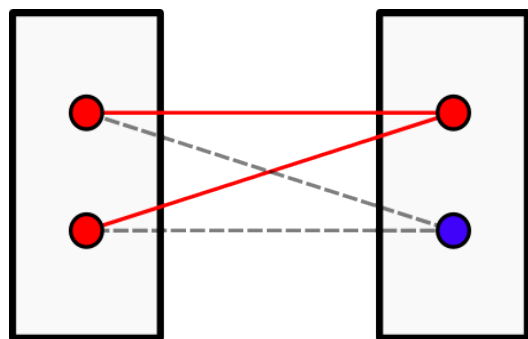
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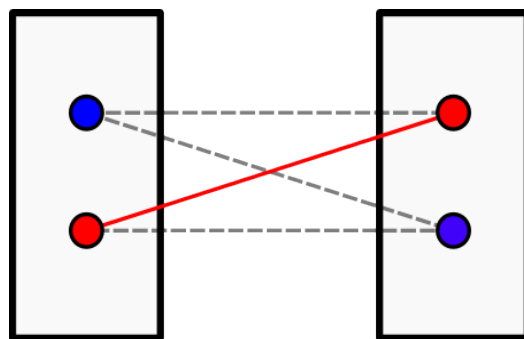
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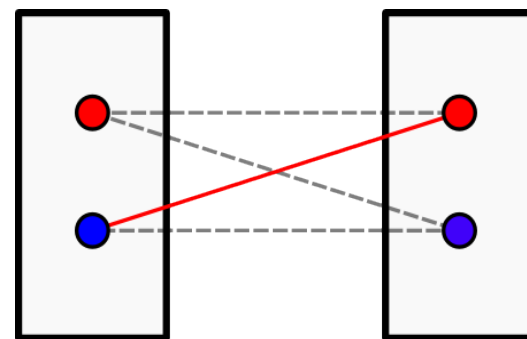
$$D(\phi) = \sum_{v \in \mathcal{V}} \min_{x_v \in \mathcal{X}_v} \theta_v^\phi(x_v) + \sum_{uv \in \mathcal{E}} \min_{(x_u, x_v) \in \mathcal{X}_{uv}} \theta_{uv}^\phi(x_u, x_v)$$



$\text{mi}[\theta^\phi]$



$\frac{\partial D}{\partial \phi} = 0$



$\frac{\partial D}{\partial \phi} \neq 0$

Proposition: ∂f – convex set

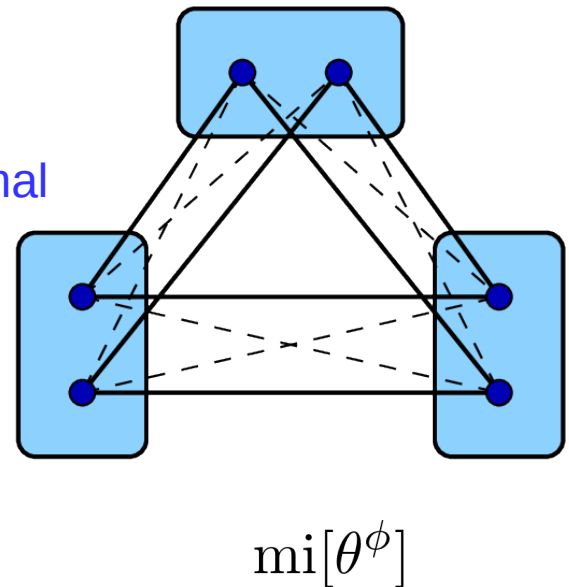
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What about this case?

- 1) There is no zero subgradient, because not integer labeling consisting of minimal labels and edges
- 2) There is a zero subgradient, because the dual is optimal
- 3) No idea



Proposition (*):

$\tilde{f}(y) = \max_{x \in D} f(x, y)$ f – differentiable on y , D – arbitrary set

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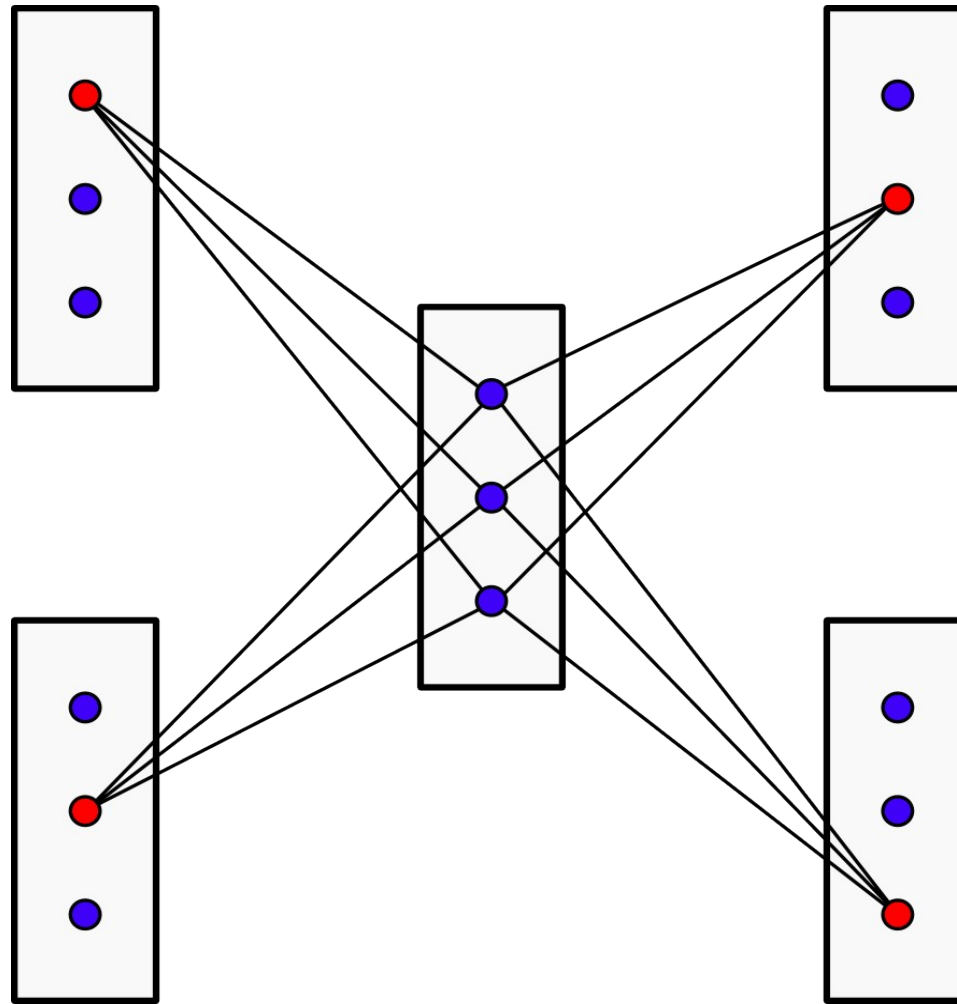
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Exercise: Based on Proposition (*) prove that

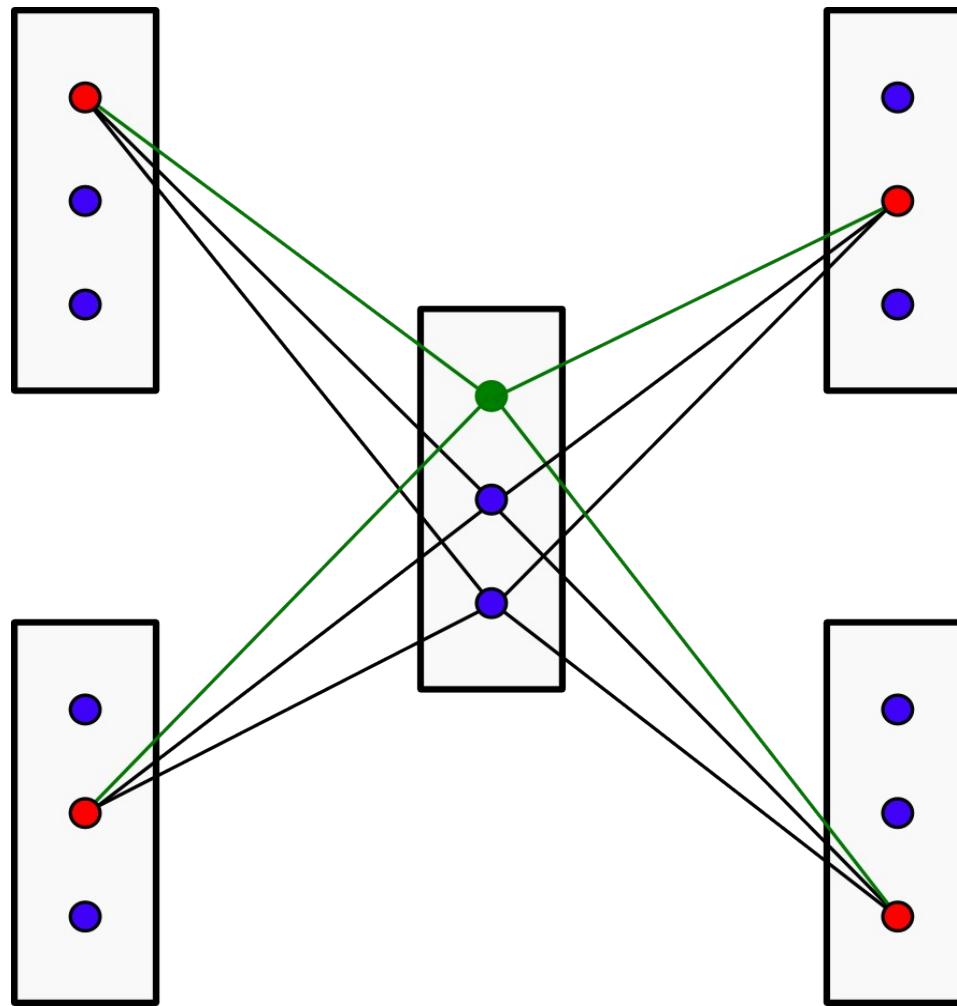
$\exists \partial D = 0$ iff there exists $\mu \in \mathcal{L}$

with $\mu > 0$ assigned to minimal edges and labels only

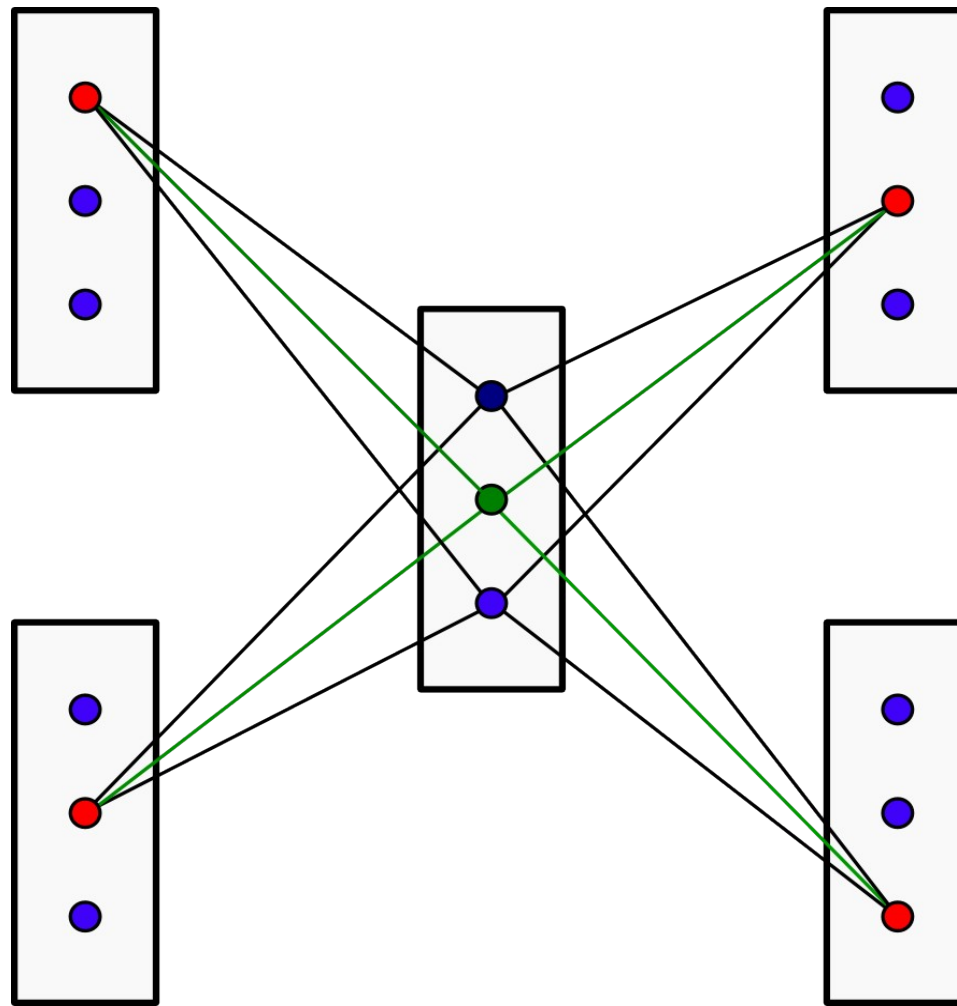
Coordinate Descent In Primal Domain: ICM Algorithm



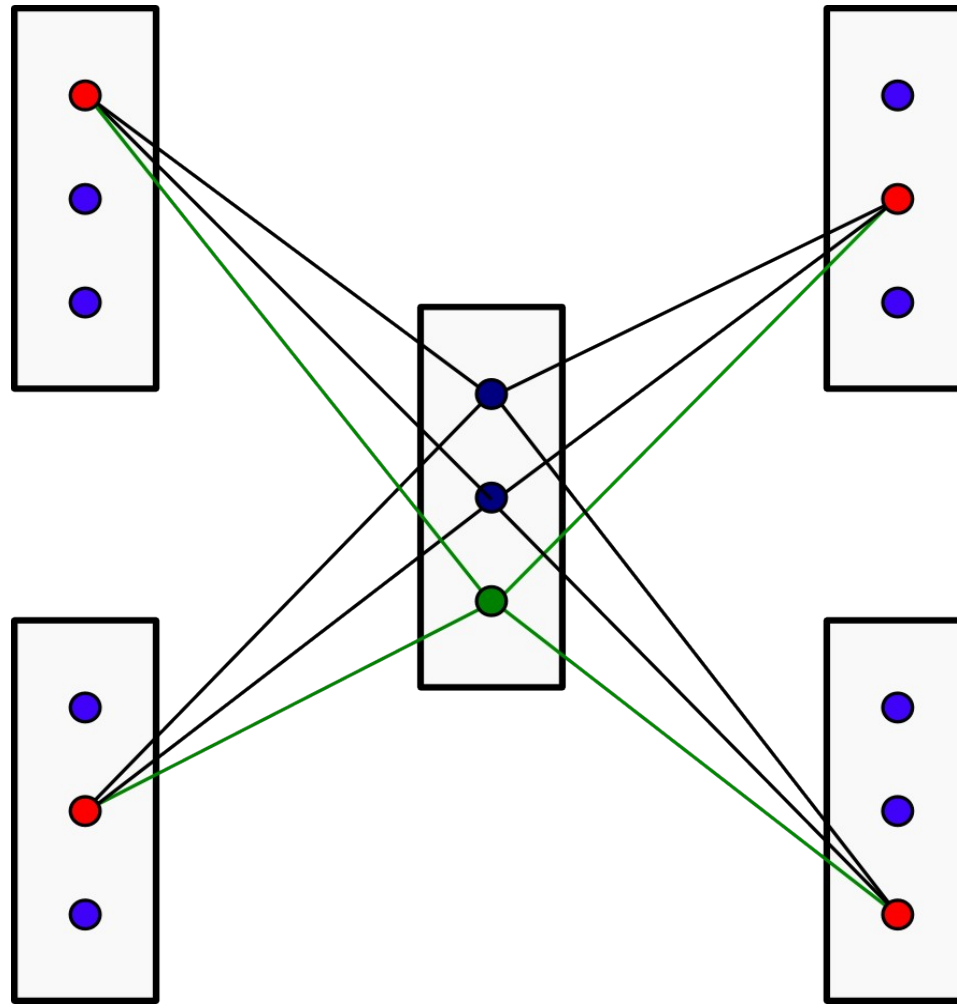
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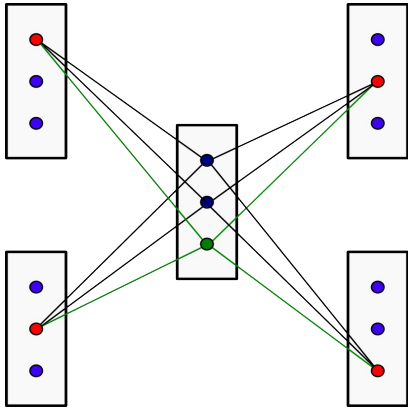
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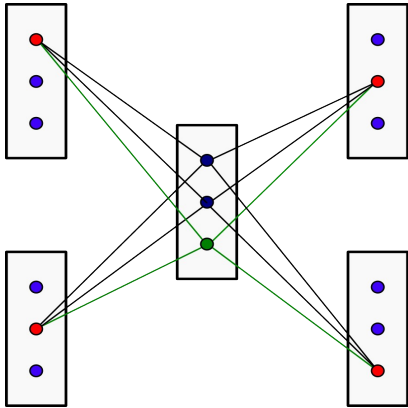


Does the ICM algorithm

- 1) converges, but only in the limit
- 2) may not converge
- 3) converges in a finite number of steps
- 4) No correct answer
- 5) I do not know

?

Coordinate Descent In Primal Domain: ICM Algorithm

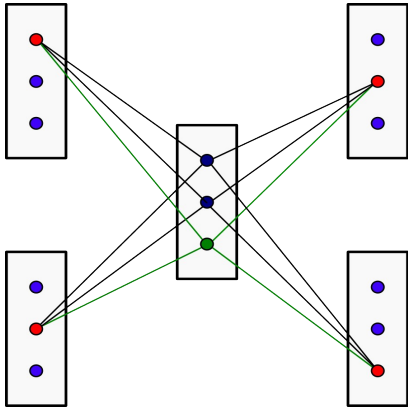


Does the ICM algorithm

- 1) find an optimal integer labeling
- 2) find an optimal relaxed non-integer labeling
- 3) there is no evidence neither for 1) nor for 2)
- 4) None is correct
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Coordinate Descent In Primal Domain: ICM Algorithm

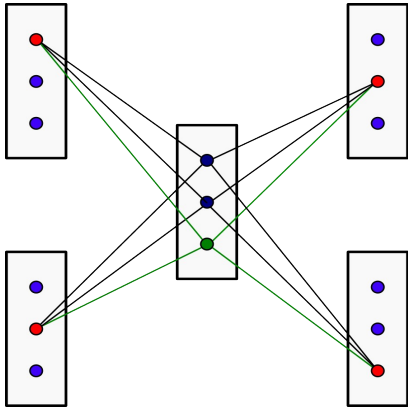


For acyclic models: does the ICM algorithm

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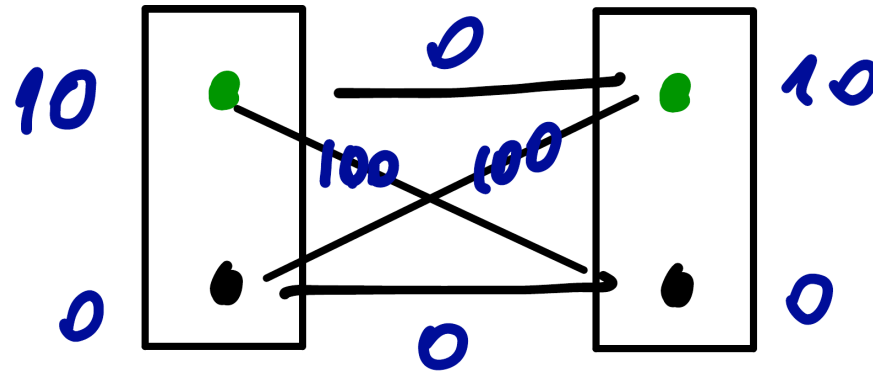
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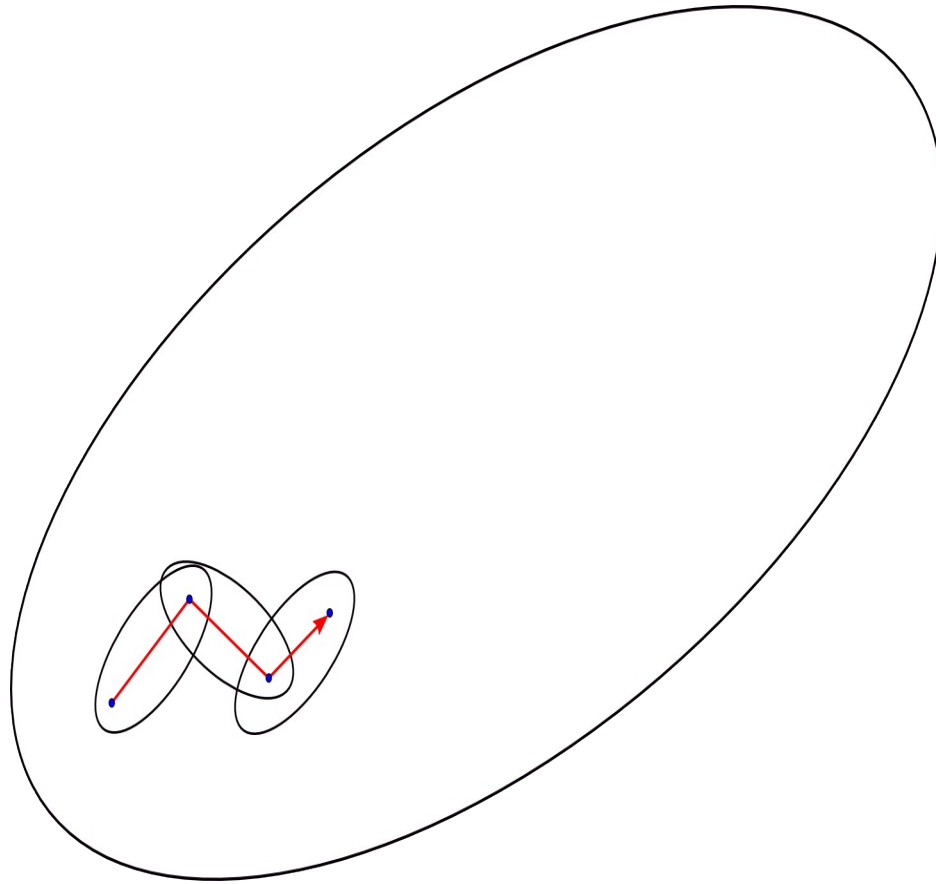
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Local search examples:

- coordinate descent
- gradient descent with line minimization
- simplex method for LPs

What is Better: Coordinate Descent In Primal or in Dual Domain? ICM or Diffusion?

Come to the next practical exercise ...